# Ecommerce Analysis

## Project Background

Deskline is an U.S.-based e-commerce company that sells office equipment, furniture, and electronics. After two years in business, the company has collected substantial data pertaining to its sales, products, and customers. However, they are unsure of how to effectively store this information, or how to translate it into actionable insights. This project establishes a new relational database environment to store current and future data, and analyzes that data to generate insights and provide recommendations for improving sales and operational performance.

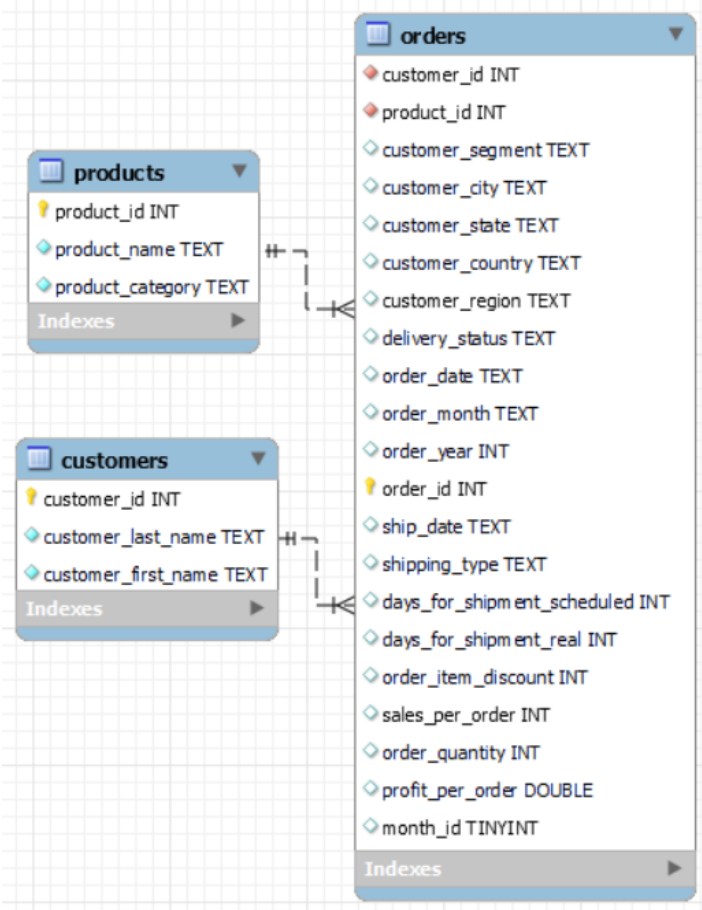
- To download an interactive Power BI dashboard, click *here*
- To view the data cleaning process in Excel and SQL, click *here*
- To view the SQL queries utilized to perform exploratory data analysis, click *here*

## Objectives

1. Transform the current data source into a structured relational database environment.
2. Identify best-performing products and categories by revenue and profitability.
3. Compare sales performance across regions and states to identify geographic strengths.
4. Analyze shipping timeliness to evaluate operational efficiency.

## Data Overview

The original dataset consisted of a single CSV file containing 21 columns and 113,271 records. To support organization, analysis, and future scalability, the data was reorganized into a relational database with three tables: **customers**, **products**, and **orders**.




Certain customer-related variables (e.g., `customer_segment`, `customer_region`) are stored in the **orders** table, as these details may vary between orders from the same customer.

# Executive Summary

put in numbers for sales The average profit per order increased from $21 to over $25 (+21%), due to a % increase in average total sales per order. Sales remained consistent from 2023–2024, while profit increased by 4.5%. Best-selling products were Office Supplies An average delivery time of 3.5 days is strong, but match shipping estimates to improve customer satisfaction and retention.

# Key Insights

## Sales Trends

- Total orders decreased by 14% from 2023–2024. However, sales revenue declined by <1%, while total profit increased by 4.5%.
- Sales and profit stability, despite fewer orders, was driven by the average sales per order increasing from $191 to $220.
- The Consumer segment accounted for 52% of total sales, followed by Corporate (30%) and Home Office (18%).
- Discounts were applied to over 94% of orders, averaging about $24 per order (11% of average order value). Deskline awarded $2,395,573 in total discounts across the two-year span.

## Product & Category Performance

- Office Supplies was the top-selling product category, accounting for over 60% of total sales.

- Variations of staples and paper were the best-selling individual products, accounting for six of the ten products with the most sales.
- All three product categories—Office Supplies, Furniture, and Technology—yielded an average profit of about $23 per order.

### Regional Sales Performance

- Deskline generated the most sales in the West region, totaling $7,420,991, or 32% of total revenue.
- The East region generated the second-highest sales (29%), followed by Central (23%) and South (16%).
- California led all states with 20% of total sales. The next-best performing state was New York, accounting for 11% of total sales.
- The average profit per order in all four regions was about $23.

### Shipping Timeliness

- Deskline frequently overestimated its delivery capabilities. The average order took 3.5 days to reach the customer, compared to the average estimated shipping time of 2.9 days.
- Inaccurate shipping estimates resulted in 55% of orders being classified as late.
- Conversely, just 41% of orders were delivered on or before the estimated delivery date.
- Average shipping time and variance were nearly identical across all regions, indicating no region-specific logistical issues.

## Recommendations

1. **Reduce discounts to improve profitability**. Decreasing the frequency of discounts from 94% to 70% of orders, and reducing the average discount from 11% to 9%, would yield an estimated $488,630 in additional profit in 2025 (+34%).
    - *Assuming order volume and average revenue per order remain constant from 2024.*
2. **Provide more accurate order shipping time estimates.** Align promised delivery times with the actual average of 3.5 days to promote customer satisfaction and retention.
3. **Prioritize Office Supplies**. Maintain sufficient inventory of high volume and profitable products, specifically staples and paper.